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Abstract—This document is a model and instructions for L^AT_EX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.

Index Terms—UAV-assisted dissemination, edge IoT, fragment-based recognition, cooperative detection, mission completion time

I. INTRODUCTION

Smart-city surveillance networks increasingly rely on distributed edge cameras, yet rapid dissemination of high-value recognition payloads remains challenging under strict mission-time constraints. In emergency pursuit scenarios, a UAV can quickly traverse a suspicious area, but each ground node is visible only for a short contact interval. If the UAV stops at each node for full payload delivery, mission latency becomes prohibitive. If it flies continuously, each node receives only partial information. This creates a fundamental trade-off between mobility efficiency and detection reliability.

This work models the UAV as a mobile data ferry that continuously broadcasts payload fragments while following an optimized path. Ground nodes accumulate confidence from partial evidence and trigger intra-cell cooperation to exchange missing fragments before the UAV leaves the region. In contrast to simplified binary-link assumptions, we explicitly model packet-level feasibility through geometry-aware air-to-ground propagation, contact-window validation, and SNR-to-BER-to-PER loss mapping.

Our approach combines (i) confidence accumulation from logical payload fragments, (ii) cooperative fragment exchange among neighboring nodes, and (iii) a Greedy Max-Coverage with Cost (GMC) trajectory heuristic to balance newly covered suspicious nodes and travel cost. The resulting framework targets earlier mission completion under realistic communication uncertainty.

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The main contributions are threefold: 1) a fragment-based confidence model with threshold-driven intra-cell cooperation; 2) a mobility-coupled packet-level PHY decision pipeline for dissemination feasibility; and 3) a practical GMC-based planning strategy that integrates candidate centroids and deterministic tie-breaking for reproducible execution. The remainder of the paper is organized as follows: Section II reviews related work, Section III presents the system model, Section IV formulates the optimization problem, Section V details the proposed method, Section VI reports evaluation results, and Sections VII–VIII discuss limitations and conclude.

II. RELATED WORK

Prior studies on UAV-assisted IoT networking have explored coverage-aware trajectory design, service-time balancing, and latency-oriented routing. These methods provide strong mobility optimization tools, but most assume monolithic payload delivery or simplified communication success models, which limits their suitability for partial-information detection tasks.

A second line of work investigates fragmented or progressive data delivery and evidence fusion. Such methods show that partial blocks can still contribute useful inference signal, and they often employ probabilistic aggregation rules to combine incomplete observations. However, these studies typically do not account for UAV mobility constraints, short contact windows, or mission-driven trajectory coupling.

A third line of work addresses distributed edge detection and cooperative inference, including threshold-triggered alerts, peer collaboration, and consensus-style decision support. Although cooperation improves robustness and can reduce false alarms, most formulations assume static infrastructure and do not jointly model mobile aerial dissemination with packet-level air-to-ground feasibility.

This paper lies at the intersection of these three directions. We jointly model UAV path planning, fragment-based confidence accumulation, and intra-cell cooperation under packet-

level PHY constraints. The key focus is minimizing mission completion time defined by early confidence attainment at a designated suspicion point.

III. SYSTEM MODEL

We consider an urban IoT surveillance network composed of a base station (BS), one UAV, and ground-node set $\mathcal{N}$. Nodes are arranged on a grid and grouped into cells to enable local cooperation. The BS identifies a suspicious subset $\mathcal{P} \subseteq \mathcal{N}$ and designates one suspicion point $n^* \in \mathcal{P}$ as the mission-completion target.

A. Network Setup and System Assumptions

The monitoring field is partitioned into cells, each containing multiple edge nodes and an optional local coordinator for message scheduling. Each node can store fragment blocks, maintain a confidence score, and exchange blocks with peers in the same cell. The BS provides mission parameters (target set, thresholds, PHY settings, and UAV start state) before flight initialization.

B. UAV Trajectory Planning Model

The UAV flies at fixed altitude with finite broadcast radius R_b and follows a waypoint sequence planned over suspicious nodes. Candidate waypoints include node positions and optional centroids. At planning step t , the next waypoint is selected by maximizing:

$$\text{score}(c) = \frac{|\text{CS}(c) \setminus \text{Covered}|}{(d(x_t, c)/v)^\alpha + \varepsilon}, \quad (1)$$

where $\text{CS}(c)$ is the suspicious-node subset covered by candidate c , $d(x_t, c)$ is travel distance from the current UAV position, and v is UAV speed. This favors candidates with high new coverage and low movement cost.

C. Data and Fragment Dissemination Model

The recognition payload is represented by K logical fragments. The UAV broadcasts fragments continuously during flight rather than pausing for full-node transfer. Each node n maintains received set $\mathcal{F}_n$ and updates confidence from observed fragments:

$$C_n = 1 - \prod_{i \in \mathcal{F}_n} (1 - p_i), \quad (2)$$

where p_i denotes the contribution of fragment i . When $C_n \geq \tau_{\text{coop}}$, node n broadcasts a manifest and requests missing fragments from peers. Mission completion is triggered when the suspicion point reaches alert confidence, i.e., $C_{n^*} \geq \tau_{\text{alert}}$.

D. Data Fusion Model

To preserve broad spatial evidence in each block, fragments are modeled by pixel-stride interleaving over an image of size $W \times H$. Fragment i contains pixels indexed by $j \equiv i \pmod{K}$, with size $s_i \approx WH/K$. This model approximates uniform information spread without requiring an end-to-end CV codec pipeline.

Each fragment contributes probability mass relative to its size:

$$p_i = 1 - (1 - C_{\text{master}})^{s_i/(WH)}, \quad (3)$$

where C_{master} is confidence under complete payload availability. Hence, receiving all fragments recovers $C_n = C_{\text{master}}$.

Confidence fusion uses the union-probability rule:

$$C_n = 1 - \prod_{i \in \mathcal{F}_n} (1 - p_i). \quad (4)$$

An equivalent log-odds representation under independence is:

$$C_n = \sigma \left(\sum_{i \in \mathcal{F}_n} \log \frac{p_i}{1 - p_i} \right), \quad (5)$$

where $\sigma(\cdot)$ is sigmoid. We use the union form in implementation for simplicity and numerical stability.

Node behavior is governed by two thresholds:

- **Cooperation threshold** τ_{coop} : trigger intra-cell manifest exchange and missing-fragment requests.
- **Alert threshold** τ_{alert} : trigger alert; mission completes when this condition is met at n^* .

Setting $\tau_{\text{coop}} < \tau_{\text{alert}}$ enables two-stage amplification, reducing direct UAV delivery burden.

E. Channel and Transmission-Rate Model

The UAV-node link follows a geometry-aware air-to-ground model with path loss, shadowing, and fast fading. Received power is:

$$P_{rx}(t) = P_{tx} - PL(d_t) - X_\sigma(t) - X_f(t), \quad (6)$$

where d_t varies with UAV motion. SNR is mapped to BER and PER:

$$\text{PER} = 1 - (1 - \text{BER})^{8L}. \quad (7)$$

Packet acceptance additionally requires contact-window validity, i.e., receiver sensitivity must hold across packet airtime. Therefore, effective rate is time-varying with geometry and channel state.

IV. PROBLEM STATEMENT

Given suspicious set $\mathcal{P}$ and designated suspicion point $n^* \in \mathcal{P}$, the UAV broadcasts fragment set $\mathcal{F} = \{1, \dots, K\}$ while moving. The objective is to minimize the time until n^* reaches alert confidence. Decision variables are: (i) UAV trajectory over waypoints and (ii) fragment transmission schedule over time.

Let $\Pi = (w_1, w_2, \dots, w_M)$ denote the UAV trajectory, where w_m is the m -th waypoint, and let $\mathcal{S} = \{(f, \tau_f)\}$ denote the schedule transmitting fragment f at time τ_f . Let $C_n(t)$ be node confidence at time t , and τ_{alert} be alert threshold. Mission detection time is:

$$T_{\text{detect}}(\Pi, \mathcal{S}) = \inf\{t \geq 0 \mid C_{n^*}(t) \geq \tau_{\text{alert}}\}. \quad (8)$$

The optimization objective is to minimize expected detection time:

$$\min_{\Pi, \mathcal{S}} \mathbb{E}[T_{\text{detect}}(\Pi, \mathcal{S})]. \quad (9)$$

The problem is subject to the following operational constraints:

$$d(w_m, w_{m+1}) \leq v_{\max} \Delta t_m, \forall m, \quad (10)$$

$$\sum_{m=1}^{M-1} \frac{d(w_m, w_{m+1})}{v} + \sum_f t_f^{\text{tx}} \leq T_{\max}, \quad (11)$$

$$\mathbf{1}_{\{n \leftarrow f\}}(t) = 1 \Rightarrow P_{rx}^{(n)}(t') \geq P_{\text{sens}}, \forall t' \in [t, t + t_f^{\text{tx}}], \quad (12)$$

$$\Pr[\text{packet error} | n, f, t] = \text{PER}(\text{SNR}_{n,f}(t), L_f), \quad (13)$$

$$\sum_{f \in \mathcal{F}_n(t)} b_f \leq B_n^{\max}, \forall n \in \mathcal{P}, \quad (14)$$

where $v_{\max}$ is maximum flight speed, $T_{\max}$ is mission deadline, P_{sens} is receiver sensitivity, L_f and b_f are packet and fragment sizes, and $B_n^{\max}$ is node buffer capacity.

Because the problem is combinatorial (discrete trajectory, coupled scheduling, and stochastic channel constraints), global real-time optimization is intractable. We therefore design a lightweight heuristic that jointly approximates mobility and dissemination decisions.

V. PROPOSED APPROACH

A. Overview

The proposed framework consists of three coordinated modules executed in a closed loop. First, the BS builds candidate waypoints and computes a GMC path for the UAV. Second, the UAV performs continuous fragment broadcasting along the path while each node updates confidence from successfully received packets. Third, nodes crossing the cooperation threshold exchange manifests and missing fragments inside each cell, accelerating confidence growth at the suspicion point.

Inputs include topology, suspicious-node set, suspicion point, threshold parameters, and PHY settings. Outputs include trajectory Π , transmission schedule $\mathcal{S}$, and mission detection time T_{detect} . We favor a heuristic design because exact optimization is computationally expensive under real-time mission constraints and random packet outcomes.

B. Algorithm

Algorithmically, the method is described at two levels. A compact end-to-end workflow orchestrates path planning, dissemination, confidence updates, cooperation triggers, and termination checks. Inside this workflow, two core procedures carry most of the novelty.

GMC Path Planner. Candidate set $\mathcal{C}$ is formed from suspicious-node positions and optional centroids. For each candidate, coverage gain and travel cost are evaluated, then scored by the GMC rule. The best candidate is appended iteratively with deterministic tie-breaking until target coverage is satisfied.

Cooperative Fragment Exchange. During flight, each successful packet updates local confidence. Once a node reaches τ_{coop} , it broadcasts a manifest, requests missing fragments, and receives peer responses under staggered scheduling. Confidence is recomputed after each exchange. Mission completion

is declared when $C_{n^*} \geq \tau_{\text{alert}}$, and the corresponding timestamp is recorded as T_{detect} .

The dissemination model uses logical payload fragments (synthetic blocks) rather than a full image-codec pipeline; thus the algorithm evaluates system-level coordination effects rather than end-to-end computer-vision preprocessing quality.

VI. PERFORMANCE EVALUATION

A. Simulation Setup

B. Metrics

C. Results

VII. DISCUSSION

VIII. CONCLUSION

ACKNOWLEDGMENT

APPENDIX